World-Model Calculus for Probabilistic Logic Networks: Evidence, Views, and Σ-Guarded Compiled Inference

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Abstract

We present a *World-Model (WM) calculus* for Probabilistic Logic Networks (PLN), formalized in Lean 4 as part of the `Mettapedia` repository.

The guiding principle is: *PLN inference is stateful posterior revision plus evidence extraction*, and "fast PLN rules" are *compiled, Σ-guarded rewrites* whose soundness is discharged relative to a declared *world-model class* (Bayesian network, factor graph, joint evidence, etc.).

We separate: (i) **evidence** as a rich algebraic object (quantale/Heyting-valued), (ii) **views** (probabilities, STVs/WTVs, typicalities) as explicit projections from evidence, and (iii) **tractable sublayers** where inference is exact relative to assumptions (e.g. d-separation). We explain how this architecture supports message passing / activation spreading, how to make approximation risk explicit, and how to integrate quantifiers via satisfying-set semantics and Goertzel-style weakness. We show that the WM calculus is not merely an abstract evidence API: it can faithfully host both Kripke and neighborhood modal semantics as concrete instantiations, recovering ordinary pointed-model truth on singleton states and yielding an evidential ensemble semantics on multisets. The neighborhood instance admits the strongest bridge: an implication-level soundness/completeness theorem connecting proof-theoretic derivability to WM consequence inequalities. A partial governance-fragment translation into neighborhood semantics is also provided.

Finally, we outline soundness and (fragmented) completeness results relative to chosen semantics.

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Status note. Lean-proven claims are those tied to explicit Lean file/theorem references (primarily §11). Other sections are research synthesis, design rationale, or roadmap. Verify against current build outputs before citing as formalization state.

Contents

1 Introduction **4**

2 The WM Kernel: States, Revision, and Queries **4**
2.1 World-model interface 4
2.2 Judgments as a proof-calculus skeleton 5
2.3 Query equivalence and Σ-guarded rewrites 6

3 Evidence vs Views **6**
3.1 Evidence as an algebraic object 6
3.2 Views as projections 7

4 No-Go Theorem **7**

5 Tractable Sublayers **7**
5.1 World-model classes and Σ side conditions 7
5.2 Variable elimination as exact query engine 8
5.3 Fast PLN rules as compiled tactics 8

6 Knuth–Skilling Bridges **8**

7 Factor Graphs: Semantics vs Control **9**
7.1 Evidence ledgers and provenance 9

8 Quantifiers in the WM Calculus **9**
8.1 Satisfying sets and weakness 9
8.2 Algorithmic Quantifier Theorem Pack (Ch. 11) 10
8.3 Higher-order quantification 10
8.4 Third-Order Quantifier Semantics 11

9 Connections **11**
9.1 Temporal/fixpoint reasoning 11
9.2 Process calculi and linear logic 11
9.3 Intensional world models 12

10 What "Sound and Complete" Should Mean Here **12**
10.1 Soundness 12
10.2 Completeness (fragmented, but meaningful) 12

11 Lean Artifact Map **13**

12 Modal Semantics Instantiations **16**
12.1 Kripke WM instance 16
12.2 Deontic Kripke consequence rules 16
12.3 Neighborhood WM instance 17
12.4 Neighborhood soundness/completeness bridge 17
12.5 Governance-to-neighborhood translation 18
12.6 Philosophical assessment 18
12.7 Limitations 19
12.8 Categorical WM Transport Layer 19
12.9 Stochastic Experiment Layer Theorem Index 20

13 Related Work **21**
13.1 Evidence logic and evidence dynamics 21
13.2 Neighborhood semantics 22
13.3 Provenance semantics and quantale-valued logics 22

14 Roadmap **22**

1 Introduction

PLN aims to support *probabilistic reasoning in open-ended domains*: uncertain facts, imperfect independence, and large-scale inference control. The historical friction is well-known:

- **Local truth-value rules are not globally complete.** Even with local evidence for A, B, C and links $A \Rightarrow B$, $B \Rightarrow C$, the *exact* evidence for $A \Rightarrow C$ cannot in general be computed without joint correlation information.
- **Independence assumptions are the real price.** When is a Markov/d-separation assumption justified? What happens when it is violated? How do you quantify drift [11]?
- **Operational factor graphs are not automatically semantics.** A message-passing graph whose factors are "deduction" or "induction" update formulas is useful for search, but semantically correct only when those factors are *compiled* from a true world-model factorization (or guarded by explicit assumptions).

The design answer:

> **Keep the semantic kernel small and honest.** World-model states revise under evidence, and queries extract evidence. Everything else—truth values, probabilities, "fast rules", message passing—are *explicit views and* Σ*-guarded tactics* whose validity is proved *relative to a declared model class.*

This aligns several foundational perspectives: Kolmogorov/Cox/Knuth–Skilling give valuation axioms and regraduation principles [6], Kohlas–Shenoy give algebraic local computation [7], and Meredith/Stay want the categorical/process structure to stay compositional.

2 The WM Kernel: States, Revision, and Queries

2.1 World-model interface

At the semantic core we posit:

- a type of world-model states $W : \mathsf{State}$,
- a type of queries $q : \mathsf{Query}$,
- an evidence type $\mathsf{Evidence}$, and
- operations *revision* and *evidence extraction*.

Conceptually:

$W \oplus \Delta$ is revision of state W by new evidence Δ, $\mathsf{evidence}(W, q) \in \mathsf{Evidence}$ is evidence for q in state W.

In Lean this is packaged as a `WorldModel` typeclass with an `EvidenceType` for states. The essential law is **additivity of evidence extraction under revision**:

$$\mathsf{evidence}(W_1 \oplus W_2, q) = \mathsf{evidence}(W_1, q) \oplus \mathsf{evidence}(W_2, q).$$

This is the "semantic spine" that makes revision and querying behave like a calculus.

For the concrete `PathMap`/`Finset` α instance, this law is Lean-proven as `pjoin_evidence_additive` in `OSLF/PathMap/PLNBridge.lean`: for disjoint stores $W_1 \perp W_2$,

$$\mathsf{evidence}(W_1 \cup W_2, q) = \mathsf{evidence}(W_1, q) \oplus \mathsf{evidence}(W_2, q),$$

where $\mathsf{evidence}(W, q) = (|W \cap q|, |W \setminus q|)$. The proof also establishes the K&S sum rule ($n^+ + n^- = |W|$), monotonicity, the Bayesian conditioning partition, and the STV connection ($\mathsf{strength} = |W \cap q|/|W|$). Axioms: standard Lean kernel only.

A `Multiset` α variant (`WorldModelBridge.lean`) resolves the idempotency obstacle: `Finset` union is idempotent ($W \cup W = W$), breaking `evidence_add` for non-disjoint pairs. `Multiset` sum is the free commutative monoid, enabling proper non-idempotent revision.

2.2 Judgments as a proof-calculus skeleton

The WM kernel yields sequent-style judgments:

- $\vdash_{\mathsf{WM}} W$ — *W is a derivable WM state*,
- $\vdash_q W \Downarrow q \mapsto e$ — *from state W, query q yields evidence e*,
- $\vdash_s W \Downarrow q \mapsto s$ — a scalar *strength view*.

This is already a *proof calculus*, but a *stateful* one: proof objects are states and evidence terms, not only formulas.

Three judgment variants track provenance at increasing precision:

1. `WMJudgment W` — context-free: "W is derivable."
2. `WMJudgmentCtx` Γ W — set-indexed: "W is derivable from states in Γ." Tightens the trust boundary.
3. `WMJudgmentMulti` Γ W — multiset-indexed: tracks multiplicities. Records if the same observation was used twice.

2.3 Query equivalence and Σ-guarded rewrites

Definition 1 (Query equivalence)**.** *Two queries* q_1, q_2 *are* evidence-equivalent *if*

$$\forall W.\ \mathsf{evidence}(W, q_1) = \mathsf{evidence}(W, q_2).$$

A "PLN inference rule" is packaged as a sound rewrite:

Definition 2 (Σ-guarded rewrite rule)**.** *A rewrite rule consists of:*

- *a side condition* Σ *(a proposition),*
- *a conclusion query* q*,*
- *a derived evidence term* $\mathrm{derive}(W)$*, and*
- *a proof that* $\Sigma \Rightarrow \forall W.\ \mathrm{derive}(W) = \mathsf{evidence}(W, q)$.

Many PLN rules rewrite only the *strength view*: they prove an identity about $\mathsf{strength}(\mathsf{evidence}(W, q))$ rather than about full evidence.

Query equivalence admits three levels of refinement, all formalized:

- `WMQueryEq` — full evidence equality (strongest).
- `WMEvidenceLE` — pointwise dominance (preorder). Antisymmetry recovers `WMQueryEq`.
- `WMViewEq` — agreement after projecting through a view function $v : \mathsf{Evidence} \to X$ (weakest).

3 Evidence vs Views

3.1 Evidence as an algebraic object

PLN needs an evidence object that encodes positive and negative support, partial truth / paraconsistency, monotone aggregation, and (in some sublayers) conjugate-prior counts.

The concrete type is:

$$\mathsf{Evidence} = (n^+, n^-) \in \mathbb{R}^{\infty}_{\geq 0} \times \mathbb{R}^{\infty}_{\geq 0}.$$

This forms:

- a **complete lattice** (coordinatewise $\leq$),
- a **commutative monoid** under $\oplus$ (coordinatewise addition: evidence aggregation),
- a **commutative monoid** under $\otimes$ (coordinatewise product: sequential composition),
- a **quantale**: $\otimes$ distributes over arbitrary joins.

All algebraic laws are Lean-proven in `EvidenceQuantale.lean`.

3.2 Views as projections

A **view** is a function $v : \mathsf{Evidence} \to X$, chosen for operational convenience:

$$\text{strength} \quad s(e) \in [0,1], \qquad \text{confidence} \quad c(e) \in [0,1], \qquad \text{ITV} \quad [l,u] \subseteq [0,1].$$

Views *must be explicit*, because they necessarily forget information. This enforcement prevents silent unsoundness.

Remark 3 (The totality gate)**.** *If* Evidence *has incomparable elements, there is no faithful order-embedding into* $\mathbb{R}$*. So a single real-valued view cannot be globally complete. This is not a bug; it is a statement about what information you are discarding.*

The view layer is parameterized by `InterpretableEvidence`:

- `BinaryContext` (Beta priors) — Bayesian strength/confidence.
- `ContinuousContext` (Normal-Gamma) — continuous inference.
- `CategoricalContext` (Dirichlet) — multinomial inference.

ITV (Indefinite Truth Value) semantics provide credible intervals on both Bayesian (Beta posterior) and Walley IDM branches [15].

4 No-Go Theorem

A foundational result proved in Lean:

Theorem 4 (No local complete link-deduction rule)**.** *There is no function*

$$f : \mathsf{Evidence}^5 \to \mathsf{Evidence}$$

that takes only local link/proposition evidence (for A, B, C *and* $A \Rightarrow B$*,* $B \Rightarrow C$*) and returns the exact evidence for* $A \Rightarrow C$ *for all world-model states.*

Interpretation: **correlations live in the joint state.** "Fast PLN rules" must be framed as admissible rewrites under assumptions; they are not globally complete. This no-go is the reason the WM calculus is *state-first*: the state carries whatever correlation structure is needed.

See `PLNJointEvidenceNoGo.lean` for the Lean proof.

5 Tractable Sublayers

5.1 World-model classes and Σ side conditions

To scale, we restrict the class of admissible states:

$$\mathcal{C} \subseteq \mathsf{State} \quad \text{(e.g. DAG-factorizing BNs, Markov random fields, etc.).}$$

Inside such a class, additional theorems (screening-off, d-separation) hold and can be compiled into admissible rewrite rules [11].

Side conditions Σ include:

- structural hypotheses (a particular DAG/factorization),
- conditional independence (from d-separation),
- positivity/nondegeneracy (to avoid division by 0),
- approximation/error ledger invariants.

5.2 Variable elimination as exact query engine

Given a discrete factor graph (or BN compiled to a factor graph), we compute unnormalized weights by variable elimination (VE) [4]:

$$Z(\text{constraints}) \;=\; \sum_{\text{consistent assignments}} \prod_{f} \phi_f.$$

In Lean, a generic VE engine is implemented for factor graphs parameterized by a semiring K, with specialization to $\mathbb{R}_{\geq 0}^{\infty}$.

Remark 5. *A `CommSemiring` assumption is* stronger *than separate `AddCommMonoid+Mul`, but it is not "making it a ring." Rings add additive inverses; semirings do not. VE is "sum of products" and needs distributivity.*

5.3 Fast PLN rules as compiled tactics

A PLN rule (e.g. deduction) becomes:

> A Σ-guarded rewrite of a hard query into smaller queries, together with a proof that the strength view of the conclusion equals the rule formula applied to the strength views of the premises.

This is a *strength-level* rewrite unless the model class supplies a law of evidence flow. The evidence for the conclusion is still obtained by querying the WM state, unless stronger results hold.

Concrete instances include BN screening-off (`PLNBNCompilation.lean`) and fast Bayesian-network rules (`PLNBayesNetFastRules.lean`).

6 Knuth–Skilling Bridges

Knuth–Skilling (K&S) "foundations of inference" axiomatize valuation schemes on lattices [6]. In our setting, K&S plays two roles:

1. **Probability as a valuation/view.** Within Boolean world sets, probabilities arise as valuations.
2. **Regraduation bridges for computation.** K&S representation theorems justify mapping an abstract valuation to a real scale where combination becomes addition:
$$\Theta(x \star y) = \Theta(x) + \Theta(y).$$
This is explicitly a **scale change**, not probability normalization.

This matters for PLN because it provides a principled bridge between abstract evidence/valuation objects and numerical algorithms; it lets the VE engine operate over K&S-valued factor graphs after regrading; and it keeps "probability" as a *view* rather than the foundational type.

The Lean formalization lives in `KnuthSkilling/Bridges/ValuationAlgebra.lean`.

7 Factor Graphs: Semantics vs Control

A key conceptual cleanup:

1. **Semantic factor graphs:** factors *are the model*. They represent a genuine factorization (BN/MRF), and VE / junction tree is exact relative to that model class.

2. **Operational factor graphs:** factors are rule schemas or control edges. Message passing is a heuristic that schedules which queries to ask. It becomes *semantics* only after compilation: each operational factor is justified by a semantic factor (or by a proven admissible rewrite under Σ).

7.1 Evidence ledgers and provenance

Message passing, activation spreading, and positive feedback loops are not disallowed. They need accountability:

- Track provenance and overlap to avoid double-counting evidence.
- Track Σ hypotheses explicitly.
- Maintain a ledger of approximation error / drift estimates.

Evidence sources can overlap (shared observations, "data incest"). A principled fix is a **partial** commutative monoid (separation algebra): $W_1 \oplus W_2$ is defined only when fragments are known independent/disjoint. If disjointness is uncertain, either keep fragments separate (delayed fusion) or fuse conservatively using bounds [15].

8 Quantifiers in the WM Calculus

PLN must support probabilistic reasoning over quantified claims.

8.1 Satisfying sets and weakness

A clean PLN-compatible semantics treats a predicate as a frame-valued satisfying set:

$$\chi_P : U \to \mathsf{Evidence},$$

analogous to a subobject classifier in a topos [9]. Universal quantification is evaluated via Goertzel's *weakness* on the diagonal relation:

$$\forall x.\, P(x) \quad \rightsquigarrow \quad \text{weakness}_{\mu}\big(\{(u,v) \mid P(u) \wedge P(v)\}\big).$$

Two distinct quantifier views are kept separate:

- **Extensional (classical) quantifiers:** "all individuals satisfy."
- **Typicality quantifiers:** "generality" measured by weakness/diagonal mass.

The separation prevents subtle category mistakes: typicality is not classical $\forall$, but it is exactly what many PLN uses want.

In the WM architecture, quantified statements are simply queries q: the world model state decides how to compute $\mathsf{evidence}(W, q)$. Different world-model classes yield different quantifier semantics: finite-domain exact, sampled approximate, or quasi-Borel/infinite.

See `PLNFirstOrder/QuantifierSemantics.lean`.

8.2 Algorithmic Quantifier Theorem Pack (Ch. 11)

`PLNFirstOrder/QuantifierAlgorithmTheorems.lean` (361 lines, 0 sorries) proves three families of results about the algorithmic fuzzy-quantifier layer:

Parameter monotonicity. The near-one, near-zero, and near-one-fraction scores are monotone in ε (`nearOne_mono_of_epsilon_le`, `nearZero_mono_of_epsilon_le`, `nearOneFraction_mono_of_epsilon_le`). Relaxing ε or PCL weakens the for-all threshold (`fuzzyForAllHolds_of_epsilon_and_PCL_relax`, `fuzzyIntervalHolds_of_wider_bounds`).

Crisp endpoints. When PCL = 1 the fuzzy for-all collapses to a fraction-equals-one test (`fuzzyForAllHolds_iff_fraction_eq_one_of_PCL_eq_one`), and the fuzzy there-exists collapses to a near-zero-fraction-equals-zero test (`fuzzyThereExistsHolds_iff_near`). At the fully crisp endpoint $\varepsilon = 0$ the for-all is equivalent to all scores being 1 (`crispForAll_endpoint_iff_allEqOne`).

Rule-4 discrepancy bounds. Rule-4 measures the gap between conjunction-based and quantification-based scoring. In the *high-gate* regime ($g \geq$ all individual scores), the discrepancy vanishes when the gate covers all witnesses (`rule4Discrepancy_bound_of_gate_high`, `rule4Discrepancy_eq_zero_of_gate_high_and_cover`). In the *low-gate* regime ($g <$ all scores), the discrepancy equals the minimum score and is bounded by g (`rule4Discrepancy_eq_minScore_of_gate_low`, `rule4Discrepancy_le_g_of_gate_low`).

8.3 Higher-order quantification

Quantifying over predicates is hard in crisp logics. Probabilistic semantics changes the game:

- **HO probabilities can be flattened.** Kyburg argues that second-order probabilities add no conceptual power [8]: they can be represented as ordinary joint distributions over expanded spaces.
- **Henkin semantics as a pragmatic HO target.** Quantify only over a chosen collection of predicates, yielding completeness metatheorems analogous to FO settings.
- **Quasi-Borel / measurable HO as the long game.** HO quantification can be interpreted as integration over function spaces—difficult, but mathematically well-defined.

8.4 Third-Order Quantifier Semantics

`PLNFirstOrder/ThirdOrderQuantifierSemantics.lean` (200 lines, 0 sorries) makes the "distribution over second-order uncertainty" layer explicit. Each domain element carries a `SecondOrderUncertainty` (a probability distribution over latent outcomes, each with an `Evidence` value); the collection over the domain forms a `ThirdOrderQuantifierModel`.

The key results:

- Universal evidence from the full domain bounds observed evidence (`forAllEvidenceUniversal_le_observ`) with equality when all elements are observed (`forAllEvidenceObserved_univ_eq_universal`).
- Universal confidence is bounded by finite-observation confidence (`confidence_universal_le_observed`).
- A *weakness bridge* connects the third-order model to the existing `forAllEval` semantics under explicit pointwise assumptions (`forAllEvalWeakness_eq_forAllEval_of_assumptions`, `forAllEvalWeakness_confidence_eq_of_assumptions`).

9 Connections

9.1 Temporal/fixpoint reasoning

Once reasoning becomes stateful (WM revision) and iterative (message passing), temporal/fixpoint logics become natural. The modal μ-calculus provides a bridge: temporal operators ("eventually", "always") can be presented as least/greatest fixpoints; PLN temporal operators embed under a suitable semantics; and fixpoint structure aligns with recurrent "anchor states" and invariant tracking. The WM side now includes an explicit Hyperseed-style consequence-closure operator with least-fixpoint semantics (`PLNWorldModelFixpointClosure.lean`) and concrete positive/negative finite fixtures (`PLNWorldModelFixpointClosureRegression.lean`).

9.2 Process calculi and linear logic

WM states can be treated as *resources* (linear logic flavor). Revision is a commutative monoidal addition of evidence/resources. Query answering is a form

of observation (a morphism out of state). Operational factor graphs become process networks, while the WM semantics provides correctness conditions.

9.3 Intensional world models

The intensional layer (`PLNIntensionalWorldModel.lean`) extends queries with sort-indexed channels: extensional, intensional-ASSOC, intensional-PAT, and mixed inheritance. This is packaged as `WorldModelSigma` — a typed variant where queries form a dependent family $\mathsf{Query} : \mathsf{Srt} \to \mathsf{Type}$.

10 What "Sound and Complete" Should Mean Here

10.1 Soundness

For a fragment $\mathcal{L}$ and semantics $[\![\cdot]\!]$:

- **Kernel soundness:** revision/query laws (additivity).
- **Rewrite soundness:** each Σ-guarded rule preserves the chosen view.
- **Approximate soundness:** rules preserve bounds or carry explicit error budget.

10.2 Completeness (fragmented, but meaningful)

Completeness is stated relative to a declared semantics and fragment:

- **Propositional:** evidence-quantale semantics vs IPL/intermediate logics.
- **First-order:** satisfying-set/weakness quantifiers (finite or Henkin FO).
- **BN fragment:** completeness relative to BN query answering (VE/junction tree).
- **Higher-order:** Henkin completeness; QBS as aspirational.

This is not a retreat. It is how serious proof theory works: different fragments have different theorems.

Scope of HOL/FOL completeness wrappers. The infinitary HOL and FOL completeness modules (`PLNWorldModelHOLInfinitaryCompleteness.lean`, `PLNWorldModelFOLInfinitaryCompleteness.lean`) provide *implication-level closure wrappers* over declared soundness/completeness assumptions of the underlying proof systems: given a pointwise implication $\varphi \Rightarrow \psi$ (or its singleton-strength equivalent), they produce a `WMConsequenceRule` witnessing $\mathsf{strength}(W, \varphi) \leq \mathsf{strength}(W, \psi)$ for all multiset states W. These are not global WM-shell completeness results; they transport proof-system guarantees into WM strength inequalities.

11 Lean Artifact Map

The main Lean artifacts (all in `Mettapedia/`):

File	Role
`Logic/EvidenceClass.lean`	Evidence typeclasses, `InterpretableEvidence`, contexts
`Logic/EvidenceQuantale.lean`	`Evidence` structure, complete lattice, quantale, views
`Logic/PLNWorldModel.lean`	WM kernel: `WorldModel`, `PLNQuery`, `WMJudgment`
`Logic/PLNWorldModelCalculus.lean`	Σ-guarded rewrites, `WMQueryEq`, strength rules
`Logic/PLNJointEvidence.lean`	`JointEvidence` (Dirichlet over worlds)
`Logic/PLNJointEvidenceNoGo.lean`	No-go theorem (Theorem 4)
`PLN/WorldModel/BayesNet/PLNBayesNetWorldModel.lean`	BN layer: CPT evidence, marginalization
`PLN/Bridges/ProbabilityTheory/BayesNet/PLNBNConfiguration.lean`	Configuration lemmas for BN
`Logic/PLNBayesNetFastRules.lean`	Compiled fast BN rules
`Logic/PLNIntensionalWorldModel.lean`	Sort-indexed queries, `WorldModelSigma`
`Logic/PLNWorldModelInstitution.lean`	Institution-style signature transport with explicit satisfaction-condition theorems
`Mettapedia/PLN/Bridges/CategoryTheory/WorldModelHyperdoctrine.lean`	Hyperdoctrine layer: query fibers, change-of-base, Beck-Chevalley transport
`Mettapedia/PLN/Bridges/CategoryTheory/WorldModelCategoricalBridge.lean`	Unifying institution transport and Beck-Chevalley transport
`Logic/PLNWorldModelITV.lean`	ITV semantics (Bayes and Walley IDM)
`Logic/PLNWorldModelITVHypercube.lean`	Interval-selector layer, typed ITV transport
`PLN/Core/PLNCanonicalAPI.lean`	One-call end-to-end endpoints
`PLN/RuleFamilies/FirstOrder/Quantifiers/Quantifiers.lean`	Quantifiers: semantics, $\exists/\forall$ duality
`PLN/RuleFamilies/FirstOrder/Quantifiers/QuantifierAlgorithmTheorems.lean`	Chain-level algorithm theorems: monotonicity, crisp endpoints, rule-4 discrepancy bounds (361 lines)

File	Role
PLN/RuleFamilies/FirstOrder/Quantifiers/ThirdOrderQuantifierSemantics.lean	Third-order quantifier semantics, weakness bridge, confidence-degradation bound (200 lines)
ProbabilityTheory/BayesianNetworks/FactorGraph.lean	Factor graphs
ProbabilityTheory/BayesianNetworks/VariableElimination.lean	VE engine
ProbabilityTheory/KnuthSkilling/Bridges/ValuationAlgebra.lean	KS valuation algebra bridge
OSLF/PathMap/PLNBridge.lean	PathMap ↔ PLN evidence bridge
OSLF/PathMap/WorldModelBridge.lean	Multiset WM instance
Logic/LP/WorldModelBridge.lean	LP least Herbrand model bridge
OSLF/Framework/QuantaleCoherence.lean	Quantale transport/coherence
Mettapedia/PLN/Bridges/Logic/WorldModelKripkeSingleton.lean	PLN WM ↔ Kripke singleton adequacy
Mettapedia/PLN/Bridges/Logic/WorldModelKripkeConsequence.lean	PLN WM ↔ Kripke consequence rules
Mettapedia/PLN/Bridges/Logic/WorldModelKripkeCompleteness.lean	PLN WM ↔ Kripke completeness theoretic bridge
Mettapedia/PLN/Bridges/Logic/WorldModelWeightedKripke.lean	PLN WM ↔ weighted Kripke, singleton adequacy
Mettapedia/PLN/Bridges/Logic/WorldModelNeighborhoodCompleteness.lean	PLN WM ↔ neighborhood completeness bridge
Mettapedia/PLN/Bridges/Logic/WorldModelNeighborhoodConsequence.lean	PLN WM ↔ neighborhood consequence families
Mettapedia/PLN/Bridges/Logic/WorldModelKripkeNeighborhoodEmbedding.lean	PLN WM ↔ Kripke–neighborhood embedding
Mettapedia/PLN/Bridges/Logic/WorldModelKripkeNeighborhoodCanonical.lean	PLN WM ↔ canonical Kripke–neighborhood
Mettapedia/PLN/Bridges/Logic/WorldModelWeightedKripkeWeighted.lean	PLN WM ↔ weighted Kripke semantics
Logic/PLNWorldModelHOLCompleteness.lean	HOL consequence-transfer wrappers + external-implication sound/complete bridge schema
Mettapedia/PLN/Bridges/Logic/WorldModelFOLCompleteness.lean	PLN WM ↔ FOL completeness: provable ⇔ singleton consequence on T-models
Mettapedia/PLN/Bridges/Logic/WorldModelSetTheoryBridge.lean	PLN WM ↔ Set theory (Z/ZF/ZC/ZFC) of the FOL implication bridge + categorical wrappers
Mettapedia/PLN/Bridges/Logic/WorldModelSetTheoryBridgeRegression.lean	PLN WM ↔ Set theory bridge endpoints for rule packaging and categorical transport

File	Role
`Languages/GF/GFWMObligationAdapter.lean`	GF syntax→WM obligation adapters aligned to `TwoSortJudgmentWMInterface`, plus faithful/non-full functor characterization
`Languages/GF/GFWMObligationAdapterRegression.lean`	GF→WM regression fixtures: direct transport, end-to-end rule consumption, and non-fullness witness
`PLN/WorldModel/Experiment/PLNWorldModelExperiment.lean`	Experiment/channel WM interface + Blackwell consequence wrapper
`PLN/WorldModel/Experiment/PLNWorldModelExperimentRegression.lean`	Experiment fixture regression
`PLN/WorldModel/Experiment/PLNWorldModelExperimentStochastic.lean`	Stochastic Blackwell utility layer + weighted/trusted-source transfer
`PLN/WorldModel/Experiment/PLNWorldModelExperimentStochasticRegression.lean`	Stochastic Blackwell/weighted-policy fixture regression
`PLN/WorldModel/Fixpoint/PLNWorldModelFixpoint.lean`	Monotone WM consequence operator, least-fixpoint closure, and threshold-soundness transport
`PLN/WorldModel/Fixpoint/PLNWorldModelFixpointRegression.lean`	Fixpoint/closure regression fixtures (rule-driven growth vs empty-rule collapse)
`Logic/PLNWorldModelHOLInfinitary.lean`	Countable $\bigwedge/\bigvee$ HOL WM instance, singleton adequacy, pointwise-implication strength transfer (291 lines)
`Logic/PLNWorldModelHOLInfinitaryCompleteness.lean`	Implication consequence-transfer wrappers: `WMConsequenceRule` from pointwise/singleton-strength (89 lines)
`Mettapedia/PLN/Bridges/Logic/WorldModelHOLInfinitary.lean`	Canonical $\bigwedge/\bigvee$ FOL WM instance, singleton adequacy, pointwise-implication strength transfer (297 lines)

File	Role
`Mettapedia/PLN/Bridges/Logic/WorldModelFOLInfinitaryCompleteness.lean`	Infinitary completeness: `WorldModelFOLInfinitary` transfer wrappers: `WMConsequenceRule` from pointwise/singleton-strength (93 lines)
`Mettapedia/PLN/Bridges/Logic/WorldModelFOLInfinitaryRegression.lean`	Regression: `WorldModelFOL` fixture endpoints (129 lines)

All files build with zero sorries and standard Lean kernel axioms only (`propext`, `Classical.choice`, `Quot.sound`).

12 Modal Semantics Instantiations

The generic WM kernel (§2) is a semantic evaluation framework: every state is derivable, and query judgments deterministically extract evidence. The question is whether this framework can host *real* modal semantics—not merely as illustration, but with genuine adequacy and proof-theoretic closure. This section describes nine Lean files (2,913 lines total, zero sorries) that answer affirmatively.

12.1 Kripke WM instance

A `PointedKripke` structure pairs a Kripke model (frame + valuation) with a designated world. A multiset of such structures forms a WM state, and evidence is extracted by counting satisfaction and refutation across the ensemble:

$$\mathsf{evidence}(\{m_1, \ldots, m_n\}, \varphi) = \big(|\{i \mid m_i \models \varphi\}|,\ |\{i \mid m_i \not\models \varphi\}|\big).$$

Here m denotes a *pointed Kripke model* (a Kripke frame with valuation plus a designated world), φ is a modal formula (`Formula` $\mathbb{N}$ in the Lean source), and W is a multiset of such pointed models. The key adequacy result recovers crisp modal truth on singleton (single-pointed-model) states:

Theorem 6 (Singleton adequacy)**.** *For any pointed Kripke model m and modal formula φ, $m \models \varphi \iff \mathsf{strength}(\{m\}, \varphi) = 1$.*

This is Lean-proven as `singleton_adequacy_strength_one` in `PLNWorldModelKripke.lean` (242 lines).

There is also a *consequence lifting* theorem: if $\forall m.\ m \models \varphi \Rightarrow m \models \psi$ (quantifying over all pointed Kripke models m), then $\mathsf{strength}(W, \varphi) \leq \mathsf{strength}(W, \psi)$ for every multiset state W (a `Multiset PointedKripke`). Singleton semantic implication is equivalent to singleton strength consequence (`pointwiseImplies_iff_singletonStrengthLE`).

12.2 Deontic Kripke consequence rules

A `PointedDeonticKripke` structure extends the above with labelled-transition-system deontic accessibility. Two modal/deontic theorems are lifted into WM consequence rules:

- **Reflexive rule:** $\mathsf{strength}(W, \mathrm{rexist}(\varphi)) \leq \mathsf{strength}(W, \varphi)$.
- **DTS obligation–permission:** $\mathsf{strength}(W, \mathrm{ob}(\varphi)) \leq \mathsf{strength}(W, \mathrm{pe}(\varphi))$.

Both are packaged as `WMConsequenceRule` instances (see `PLNWorldModelKripkeConsequence.lean`, 206 lines). This turns modal/deontic theorems into *operational WM consequence rules* rather than leaving them as abstract semantic lemmas.

12.3 Neighborhood WM instance

The neighborhood instantiation is structurally parallel to the Kripke one: a `PointedNeighborhood` pairs a neighborhood model with a world, and evidence counts neighborhood-satisfaction across a multiset ensemble. Singleton adequacy and consequence lifting hold by the same schema (`PLNWorldModelNeighborhood.lean`, 274 lines).

12.4 Neighborhood soundness/completeness bridge

The strongest result in this batch connects proof-theoretic derivability to WM consequence. Given a neighborhood proof system $\mathcal{S}$ and a frame class C equipped with imported `Sound` and `Complete` instances:

Theorem 7 (Implication-level bridge)**.** $\mathcal{S} \vdash (\varphi \to \psi) \iff \mathrm{singletonStrengthLE}_C\, \varphi\, \psi$.

From provable implication, one obtains a multiset WM strength inequality over all states whose pointed models lie in the target frame class.

Concrete instantiations are given for E, EMN, EMT (reflexive), and ED (serial). See `PLNWorldModelNeighborhoodCompleteness.lean` (315 lines).

An analogous Kripke proof-theoretic bridge is proved in `PLNWorldModelKripkeCompleteness.lean` (238 lines): for a normal modal logic with Kripke completeness on a frame class C, $\mathcal{S} \vdash (\varphi \to \psi) \iff \mathrm{singletonStrengthLE}_C\, \varphi\, \psi$ on Kripke WM states. Both semantic regimes now stand at the same metatheoretic level.

A canonical Kripke-to-neighborhood *representation bridge* is proved in `PLNWorldModelKripkeNeighborhood` (365 lines): each Kripke frame is embedded as a neighborhood frame via the standard presentation $\mathcal{N}(w) = \{X \mid R[w] \subseteq X\}$, satisfaction is preserved, and singleton/multiset WM strength agrees under the embedding. This makes the "unification" thesis concrete: the two instantiations are not merely parallel—they are connected by a proven transport.

These results are *not* completeness of the generic WM calculus; they *are* implication-level proof-theoretic consequence-transfer theorems for instantiated modal WM semantics. The distinction matters:

Remark 8. *The generic WM shell remains a* ***semantic evaluation framework****, not a standalone logic with a deep state-derivability proof theory. Every state is trivially derivable (`WMJudgment.axiom`); query judgments are deterministic (`WMQueryJudgment`). The modal instantiations make the shell* useful *by giving it strong imported semantics.*

12.5 Governance-to-neighborhood translation

The file `PLNWorldModelNeighborhoodConsequence.lean` (613 lines) packages neighborhood consequence rules and provides a recursive governance-fragment translation:

Governance modality	**Neighborhood image**
rexist(φ)	$\Box\varphi$
obligatory(φ)	$\Box\varphi$
permitted(φ)	$\Diamond\varphi$
forbidden(φ)	$\Box\neg\varphi$
optional(φ)	$\Diamond\varphi \wedge \Diamond\neg\varphi$

The translation handles propositional connectives recursively and returns `none` for μ/ν formulas (explicitly outside the current fragment). All governance action labels on box/diamond operators collapse to a single neighborhood $\Box/\Diamond$. This is a **disciplined interoperability layer for a fragment**, not a full faithful embedding of the governance modal-μ language.

Modal family theorems are proved for K, C (conjunction closure), McKinsey ($\Box\Diamond\varphi \leq \Diamond\Box\varphi$), and Makinson, with concrete frame class instantiations (EK, EMK, EC, EMC). Endpoint block theorems collect all governance-side inequalities.

12.6 Philosophical assessment

Three points emerge from this development.

A. The WM calculus is becoming a semantic unifier. The WM layer is broad enough to host crisp Kripke semantics on singleton states, evidential aggregation over multiset ensembles, and neighborhood semantics with proof-theoretic closure. The right philosophical role for the WM calculus is not "this is the final logic of reasoning," but rather: *this is a common evidential/query-calculus interface in which multiple semantic regimes can live.*

B. Multisets of pointed models are an evidential ensemble semantics. A multiset of pointed models reads as a bag of candidate scenarios, source-indexed hypotheses, or observations with multiplicity. Then $\mathsf{strength}(W, \varphi)$ becomes "how much of the evidential ensemble supports φ?" rather than "is φ true at the actual world?" This makes modal semantics relevant to PLN-style evidence reasoning: it is no longer only about one actual world with accessibility; it is *structured evidence over candidate worlds.*

C. Neighborhood semantics is the more native long-term fit. Kripke is the natural first adequacy witness. But neighborhood semantics supports weaker and non-normal modal behavior (proof systems E, EMN, EMT, ED, etc.), and the soundness/completeness bridge transports those proof systems

into WM strength inequalities. This is directly relevant if the long-term aim includes governance, evidence, non-normal modalities, and context-sensitive support.

12.7 Limitations

For clarity on what is *not* proved:

1. The **generic WM calculus does not have a nontrivial global proof theory.** The modal instantiations supply semantics; they do not give the shell itself a deep derivability structure.

2. The **governance translation is partial and lossy**: μ/ν formulas map to `none`, and all governance action labels collapse. It is a bridge for a fragment.

3. The **base multiset semantics is uniform-count**: each pointed model contributes one count. A **weighted/source-aware variant** is now formalized (`PLNWorldModelKripkeWeighted.lean`: source-labelled weighted states, trusted-source gating, weight-1 specialization bridge) and extended to stochastic channels (`PLNWorldModelExperimentStochastic.lean`: Blackwell utility transfer under arbitrary source weights). The remaining gap is **dependence and correlation correction**: the current weighted semantics treats sources as independent.

12.8 Categorical WM Transport Layer

Beyond modal instantiations, the WM layer now has an explicit categorical transport contract:

- **Institution transport** over typed WM signatures: `satisfactionCondition_evidence`, `satisfactionCondition_strength`, `satisfactionCondition_strengthWith`, `satisfactionCondition_confidence`, and derived `WMInstitution.sat_condition_strength` (`PLNWorldModelInstitution.lean`).

- **Hyperdoctrine transport** over typed query fibers with change-of-base and Beck-Chevalley: `beckChevalley_transport_exists` (`PLNWorldModelHyperdoctrine.lean`).

- **Unified endpoint**: `EndpointStatement`, `institution_beckChevalley_endpoint`, `endpointContract_of_hyperdoctrine` (`PLNWorldModelCategoricalBridge.lean`), which packages institution-level satisfaction transport and hyperdoctrine-level Beck-Chevalley transport in one theorem statement.

- **Concrete Ch.8 fixture**: `institution_beckChevalley_endpoint_id_fixture` (`PLNWorldModelCategoricalRegression.lean`), instantiating the unified endpoint on an identity pullback square.

- **Aligned consequence-transfer wrappers**: categorical-contract variants are exposed for the two-sort-calculus→WM bridge and for HOL/FOL WM consequence transfer (`PLNWorldModelTwoSortBridge.lean`, `PLNWorldModelHOLCompleteness.lean`, `PLNWorldModelFOLCompleteness.lean`). The FOL layer now includes a full implication bridge `T |- (phi -> psi) <-> singletonConsequenceOnTheory T phi psi`, while HOL exposes a proof-system-agnostic sound/complete schema (`externalImplication_iff_singletonConsequence_of_sound_complete`).

- **Set-theory specialization**: the FOL bridge is now instantiated for set language $\mathcal{L}_{\text{set}}$ (`L_set`) and `Z/ZF/ZC/ZFC` (`PLNWorldModelSetTheoryBridge.lean`), including `provable_imp_iff_singletonStrengthLEOnTheory`, multiset consequence transport, and categorical-contract wrappers.

- **OSLF→NTT→WM canonical endpoint**: presheaf/subobject modal transport and formula-level graph witness transport are exposed via `modalSubobject_commDi_bc_graph_e` `mettaFull_modalSubobject_commDi_bc_graph_endpoint`, `oslf_formula_ntt_graph_triangle_catego` The closure layer now also exposes an evidence-first endpoint `semE_fragment_formulaCategoricalEndpoi` plus regression projection `semE_fragment_categorical_endpoint_step_fixture`, so the institution/Beck-Chevalley endpoint and one-step semE evidence closure are consumable together before any strength-view lift assumptions. Direct fixtures include regression fixtures `formula_graph_triangle_fixture` and `formula_graph_triangle_categorical_fixture` (`OSLF/Framework/ModalSubobjectBridge.lean`, `OSLF/Framework/OSLFNTTWMBridge.lean`, `OSLF/Framework/OSLFNTTTheoryClosure.lean`, `OSLF/Framework/OSLFNTTWMBridgeRegression.lean`).

- **GF→WM obligation adapters**: GF syntactic transports are consumable on the same `TwoSortJudgmentWMInterface` used by Pure-kernel bridge wrappers, via `gfSyntaxHom_to_wmStrengthObligation`, `wmConsequenceRuleOn_of_gfSyntaxHom_viaPu` `wmConsequenceRuleOn_of_fregeStrong_viaPureInterface`, with category-side characterization theorem `syntaxToWMFunctor_not_full_of_constant_evidence` (`Languages/GF/GFWMObligationAdapter.lean`, `Languages/GF/GFWMObligationAdapterRegression.le`

Theorem-index scope split (FOL vs HOL). For readers tracking theorem strength precisely: the FOL side is now a full implication bridge over theory-models `(T |- (phi -> psi)) <-> singletonConsequenceOnTheory T phi psi` and is consumable through concrete WM rule constructors (`wmConsequenceRuleOn_of_provable_imp` with regression fixture `fol_provable_bridge_rule_singleton_fixture`). The HOL side, by contrast, is intentionally schema-level in this module: it provides proof-system-agnostic transfer templates pending concrete imported HOL sound/complete instances. File names currently retain the `*Completeness` suffix for import stability, but this layer should be read as consequence-transfer closure wrappers.

For executable regression coverage, the script-level target is `scripts/check_wm_categorical.sh`.

12.9 Stochastic Experiment Layer Theorem Index

Content	Lean theorem(s)	Module
Blackwell factorization utility transport and optimal-value monotonicity	`expectedUtility_eq_of_blackwellFactor`, `optimalValue_mono_of_blackwellFactor`, `optimalValueWeighted_mono_of_blackwellFactor`, `optimalValuePolicy_mono_of_blackwellFactor`, `optimalValue_trustedGate_mono_of_blackwellFactor`	`PLNWorldModelExperimentStochastic`
Concrete finite stochastic fixture endpoints	`fixture_expectedUtility_eq_blackwell`, `fixture_optimalValue_mono`, `fixture_trustedPolicy_optimalValue_mono`	`PLNWorldModelExperimentStochastic`

Script-level regression target for this layer: `scripts/check_wm_experiment.sh`.

13 Related Work

13.1 Evidence logic and evidence dynamics

The closest prior work to the WM calculus is the *evidence logic* program of van Benthem, Fernández-Duque, and Pacuit [13, 12]. Their *evidence models* $\mathcal{M} = \langle W, E, V \rangle$ assign to each state a family of evidence sets (neighborhoods). The modal operator $\Box\varphi$ means "the agent has evidence for φ" and is non-normal: $\Box\varphi \wedge \Box\psi$ need not imply $\Box(\varphi \wedge \psi)$, because evidence pieces may be individually consistent but jointly contradictory. Belief $B\varphi$ is then *derived* from evidence via a scenario-combination policy. This architecture—evidence as a first-class structure, queries evaluate against it, belief/views are derived—is the same pattern as the WM calculus.

Van Benthem and Pacuit [14] extend this to *dynamic* evidence management: new evidence is added, old evidence revised, and belief updates accordingly. Their "revise evidence, then extract belief" pipeline is the closest prior analogue to our $W \oplus \Delta$ followed by $\mathsf{evidence}(W, q)$.

The key difference is the evidence type. In evidence logic, evidence consists of *sets of worlds* (Boolean neighborhoods). In the WM calculus, evidence is *quantitative*: a pair $(n^+, n^-) \in \mathbb{R}^\infty_{\geq 0} \times \mathbb{R}^\infty_{\geq 0}$ carrying positive and negative support counts. This makes the WM calculus oriented toward PLN-style numerical inference rather than epistemic modal reasoning. But the structural insight—that evidence should be algebraically rich, not collapsed to a truth value—is shared.

A further distinction: the WM calculus uses *commutative* additive revision ($W_1 \oplus W_2 = W_2 \oplus W_1$), reflecting that evidence aggregation is order-independent. The evidence-dynamics literature [14] includes richer non-commutative update operators (public announcement, evidence addition with priority) where revision order matters. Extending the WM calculus with order-sensitive revision is future work.

Fitting [5] formalizes a complementary perspective: BLE (Basic Logic of Evidence) allows evidence to be *incomplete or contradictory*, with a modal embedding into KX4 where $\Box X$ reads as "there is implicit evidence for X." This

structurally parallels the WM's evidence pairs (n^+, n^-) where both components can be nonzero, though Fitting's framework is qualitative (justification terms) rather than numeric counts.

13.2 Neighborhood semantics

The neighborhood WM instances (§12.3–12.4) build directly on the neighborhood semantics tradition surveyed by Pacuit [10]. Neighborhood frames generalize Kripke frames and provide semantics for non-normal modal logics, including the monotone logics E, EMN, EMT, and ED for which we prove soundness/completeness bridges into WM consequence.

13.3 Provenance semantics and quantale-valued logics

The WM's evidence quantale and the additivity law $\mathsf{evidence}(W_1 \oplus W_2, q) = \mathsf{evidence}(W_1, q) + \mathsf{evidence}(W_2, q)$ (where $+$ is coordinatewise addition on evidence) place it in the tradition of *semiring provenance semantics*. Dannert and Grädel [3] show that semiring-valued model checking for guarded logics yields algebraic provenance values tracking *how and why* a formula holds—not merely whether it does. The WM's `WMQueryEq` (certified rewrite producing the same evidence) and `evidence_add` (compositional aggregation) are instances of this pattern.

Abramsky and Vickers [1] establish quantales as algebraic semantics for logics and processes, unifying lattice order, monoidal composition, and infinitary join distribution. This motivates the choice of a quantale (rather than a bare lattice or semiring) for Evidence: quantale structure is exactly what makes $\otimes$-distributing-over-joins a theorem rather than an assumption. Bacci et al. [2] develop propositional logics interpreted in the Lawvere quantale with proof systems and decidable completeness results, providing a modern precedent for quantale-valued views.

14 Roadmap

The most valuable next theorems:

1. **Shell-clarifying theorems.** Make the meta-status of the generic WM core explicit: (a) *triviality*: $\forall W, \vdash_{\mathsf{WM}} W$; (b) *determinism*: if $\vdash_q W \Downarrow q \mapsto e_1$ and $\vdash_q W \Downarrow q \mapsto e_2$, then $e_1 = e_2$.

2. **Weighted / source-labeled multisets.** Replace plain multisets with source labels, provenance classes, or weighted multiplicities, so the ensemble semantics models source reliability and dependence.

3. **Independence-to-screening-off bridge.** Prove that d-separation implies the screening-off equalities required by PLN deduction rules, under explicit positivity.

4. **VE correctness as a WM-strength rule.** Show that VE computes the same strength view as querying the BN world model.

5. **Quantifier rewrite rules with explicit approximation ledgers.** Hook satisfying-set quantifiers into the rewrite layer with soundness under stated domain hypotheses (finite, sampled, exchangeable, etc.).

6. **Operational integration (MORK).** Implement message passing as inference control that proposes Σ-guarded rewrites, while the WM layer remains the single source of semantic truth.

Acknowledgments

This paper compiles ideas emerging from ongoing collaboration around PLN, MeTTa, and the Lean formalization in `Mettapedia`, with conceptual inspiration from: Knuth–Skilling foundations of inference, valuation algebras, belief propagation / junction trees, topos semantics of quantifiers, and higher-order process calculi.

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